

HE2003 Econometrics I

Tutorial 3

02/09/2024

Question 1

We discussed the selection bias/survivorship bias in class. Can you give some other examples of this bias?

Many good examples of “selection bias” or “survivorship bias” can be found. The following examples are especially close to the “return-to-education” example:

$$investment_i = \beta_0 + \beta_1 experience_i + u_i, \quad i = 1, \dots, n$$

where $investment_i$ refers to person i 's investment gain in the financial market, $experience_i$ refers to years of investment experience, and the error term u_i includes i 's innate ability. Therefore, u_i will be positively correlated with the explanatory variable $experience_i$ as those with higher innate ability are more likely to survive in the market and gain years of experiences.

Question 1

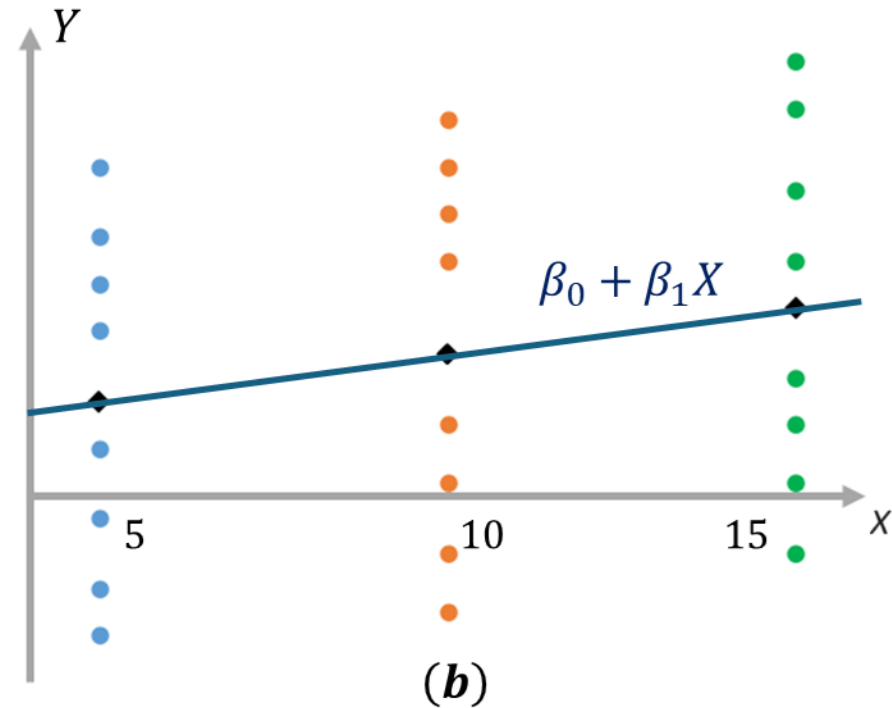
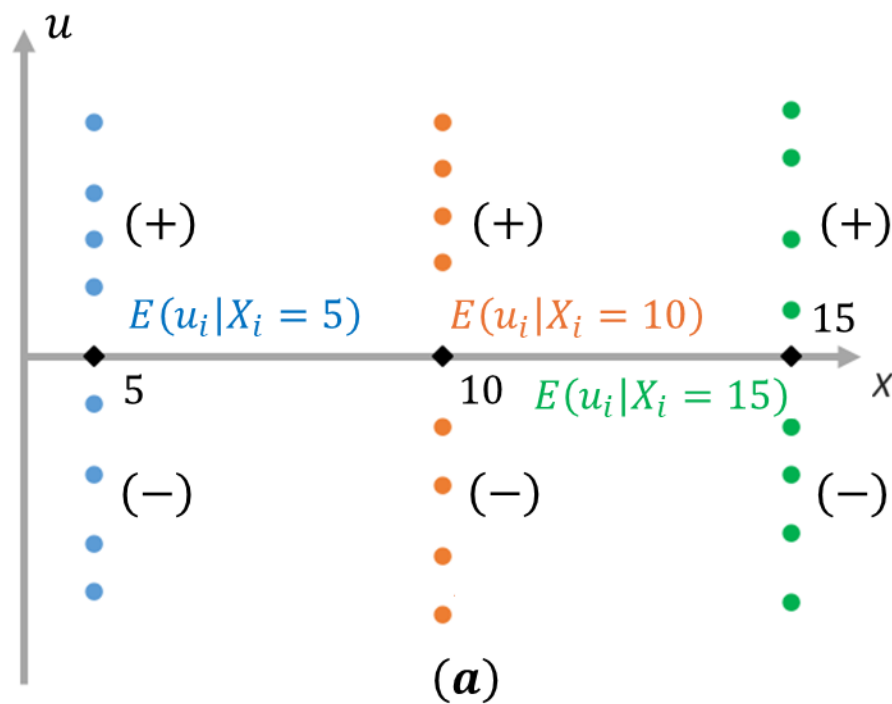
Similarly,

$$health_i = \beta_0 + \beta_1 gym_i + u_i, \quad i = 1, \dots, n$$

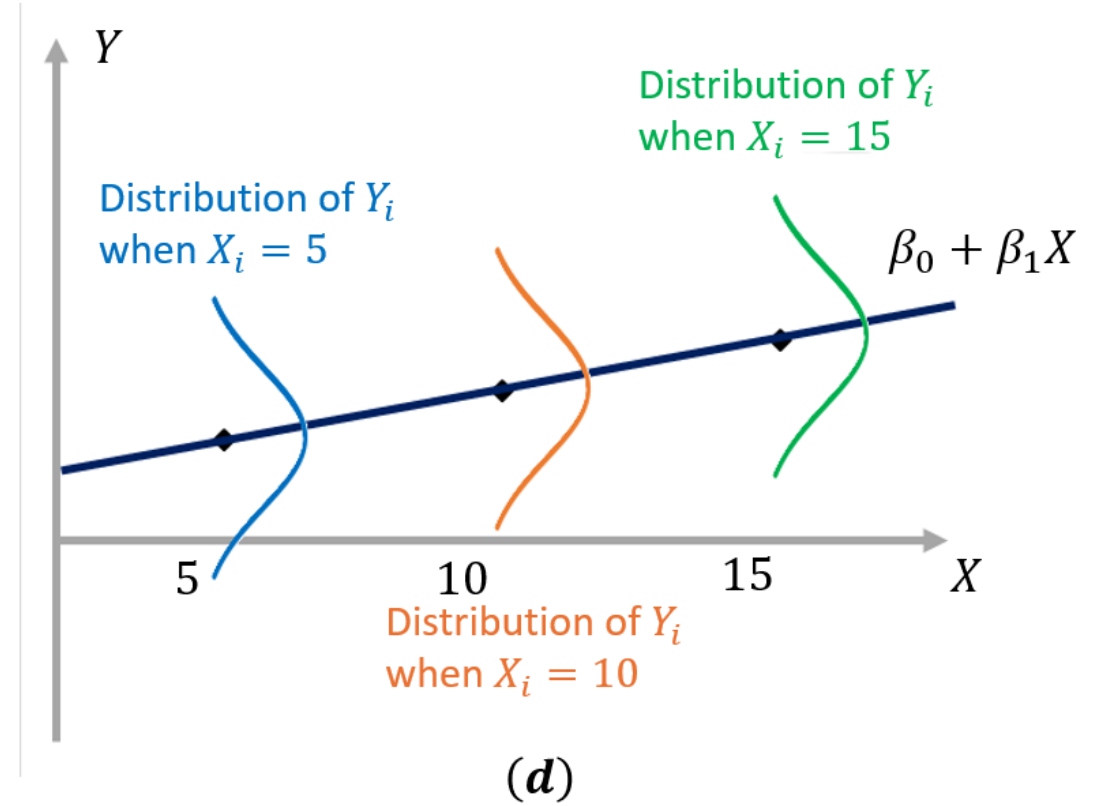
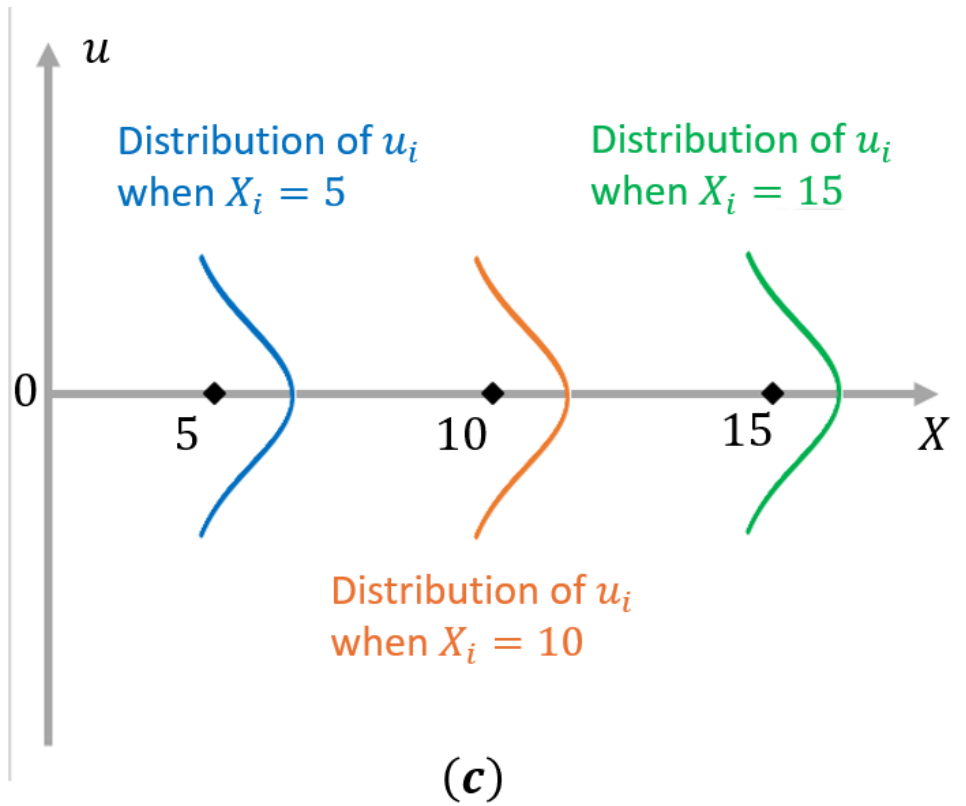
where $health_i$ refers to person i 's health condition, gym_i refers to hours of gym exercises, and the error term u_i includes i 's innate ability. Here, u_i may be positively correlated with gym_i because on average, people with higher innate physical ability (or mental ability such as endurance) are more likely to survive hard exercises and continue the training.

Question 2

Explain the meaning of the OLS Assumption 1 by using diagrams. Give example where $E[u] = 0$ but $E[u|X] \neq 0$ by using diagrams.



Question 2



Question 2

Figure 1(a) shows the **relationship between the error term u and the explanatory variable X** . When OLS Assumption 1 holds, $E[u_i|X_i = 5] = 0$. Therefore, at the point “5”, some error terms are positive (+), and some error terms are negative (−). But the average of these error terms is equal to zero at the point “5”. Similarly, OLS Assumption 1 also requires that $E[u_i|X_i = 10] = 0, E[u_i|X_i = 15] = 0, \dots$. That is, the average of the error terms are also equal to zero at the points “10”, “15” and other points of X .

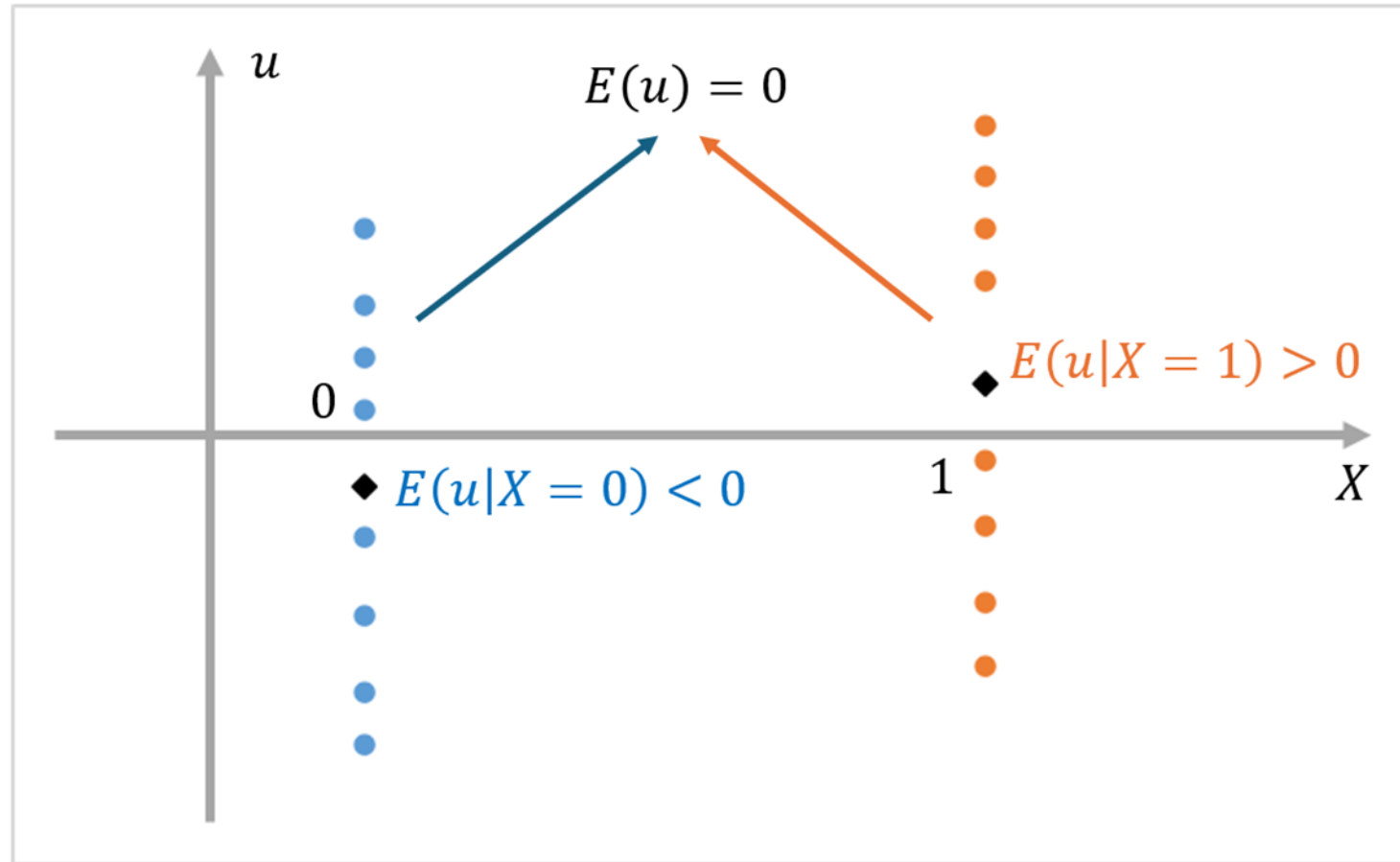
Figure 1(b) shows the **relationship between the dependent variable Y and the explanatory variable X** . This figure is obtained by combining Figure (a) with the population regression line, which is $\beta_0 + \beta_1 X$ (Note that each Y_i can be obtained by combining the **unsystematic part u_i and the systematic part $\beta_0 + \beta_1 X_i$**). As can be seen in Figure (b), at the point “5”, some values of Y_i are above the population regression line, and some values of Y_i are below the population regression line. But the average of these Y_i has to be exactly on the population regression line (according to OLS Assumption 1). Similar requirement for Y_i at the points “10”, “15”, etc.

Question 2

Figure 1(c) has the same meaning as Figure (a). The only difference is that Figure (c) replaces the data points in Figure (a) with the **distributions of u_i** at $X_i = 5, 10, 15$. Notice that for each value of X_i , the density curve of u_i is highest at zero, which means that the data point of u_i are most likely to be observed around zero. Figure 1(d) has the same meaning as Figure (b). The only difference is that Figure (d) replaces the data points in Figure (b) with the distributions of Y_i at $X_i = 5, 10, 15$.

To give an example where $E[u] = 0$ but $E[u|X] \neq 0$, for simplicity we can consider the case where X only takes two values such as 0 and 1. Then, we can show in the diagram that even if $E[u|X = 0]$ is negative while $E[u|X = 1]$ is positive, it is still possible for $E[u]$ to be equal to zero as it is the overall average over the two cases (both $X = 1$ and $X = 0$).

Question 2



Question 3

Suppose we want to find out if there exists difference in average wages between graduates from the School of Social Sciences (SSS) and graduates from the School of Humanities (SOH) using the linear regression model with an indicator variable ($X = 1$ or 0).

- (a) Write down the linear regression model for this problem. Define the dependent variable, explanatory variable, and error term in your model.
- (b) Write down the corresponding OLS Assumption 1. Explain in what case the OLS Assumption 1 is likely to hold/unlikely to hold.
- (c) Now suppose that the OLS Assumption 1 does hold, explain the specific meaning of the parameters β_0 and β_1 using conditional expectation.
- (d) Illustrate the meaning of β_0 and β_1 by using a diagram.

Question 3

(a) Write down the linear regression model for this problem. Define the dependent variable, explanatory variable, and error term in your model.

We may define the following linear regression model:

$$Y_i = \beta_0 + \beta_1 X_i + u_i, \quad i = 1, \dots, n$$

where Y_i is the wage of the graduate i , X_i is the binary regressor with $X_i = 1$ if individual i graduated from SSS and $X_i = 0$ if individual graduated from SOH, and u_i is the error term, which includes all the other factors that may affect i 's wage.

Question 3

(b) Write down the corresponding OLS Assumption 1. Explain in what case the OLS Assumption 1 is likely to hold/unlikely to hold.

The OLS Assumption 1 can be written as

$$E[u|X] = E[\text{innate ability} | \text{school type}]$$

In particular, this implies that $E[\text{innate ability} | \text{SSS}] = E[\text{innate ability} | \text{SOH}]$, that is, the average ability of SSS graduates is the same as that of SOH graduates. For example, if the students are randomly admitted to the two programs (SSS and SOH), then we can expect that the OLS Assumption 1 to hold. However, in practice the admission is not based on random assignment, so that there may exist difference in average abilities between SSS and SOH.

Question 3

(c) Now suppose that the OLS Assumption 1 does hold, explain the specific meaning of the parameters β_0 and β_1 using conditional expectation.

By the derivations in the lecture note, we have in general,

$$E[Y_i|X_i] = E[\beta_0 + \beta_1 X_i + u_i|X_i] = E[\beta_0|X_i] + E[\beta_1 X_i|X_i] + E[u_i|X_i] = 0 \text{ due to OLS assumption 1} \\ \Rightarrow E[Y_i|X_i] = \beta_0 + \beta_1 X_i$$

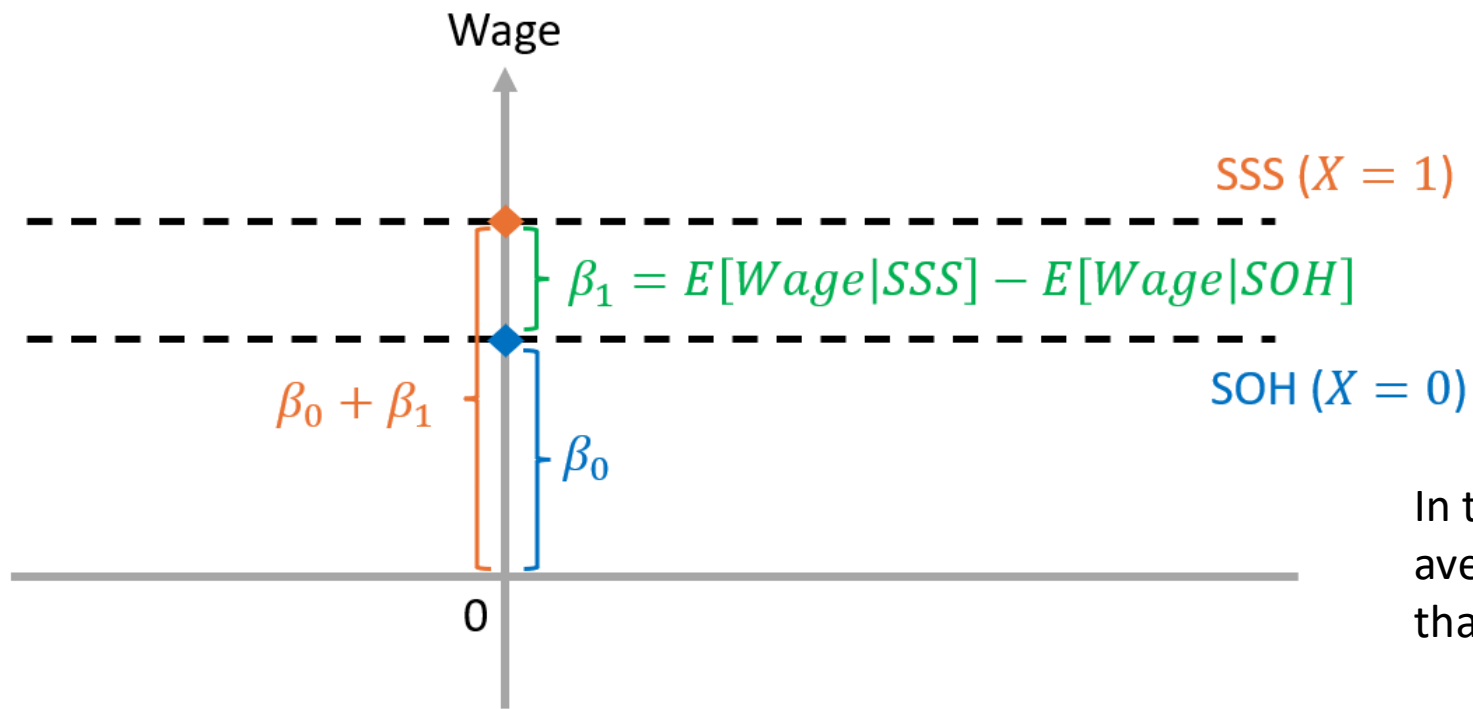
Then, for $X_i = 1$ or 0 , we have:

$$E[Y_i|X_i = 1] = \beta_0 + \beta_1 \text{ and } E[Y_i|X_i = 0] = \beta_0 \\ \Rightarrow \beta_1 = E[Y_i|X_i = 1] - E[Y_i|X_i = 0]$$

Therefore, β_0 corresponds to the average wage of the graduates from SOH. β_1 corresponds to the difference in average wages between graduates from SSS and SOH. Notice that this is a model with binary regressors, so β_1 is **not interpreted as a slope or rate of change**.

Question 3

(d) Illustrate the meaning of β_0 and β_1 by using a diagram.



In this case, we assume that the average wage of SSS is higher than that of SOH ($\beta_1 > 0$)

Question 4

Let $\hat{\beta}_0$ and $\hat{\beta}_1$ be the estimated intercept and slope from the OLS regression of the linear regression model: $Y_i = \beta_0 + \beta_1 X_i + u_i$, $i = 1, \dots, n$. Let $\tilde{\beta}_0$ and $\tilde{\beta}_1$ be the estimated intercept and slope from the OLS regression of $\tilde{Y}_i$ on $\tilde{X}_i$, where $\tilde{Y}_i = 10 \times Y_i$ and $\tilde{X}_i = 10 \times X_i$, $i = 1, \dots, n$.

Show the relation between the formula of $\tilde{\beta}_1$ and that of $\hat{\beta}_1$, and the relation between the formula of $\tilde{\beta}_0$ and that of $\hat{\beta}_0$

(Hint: Write down the formulas of the OLS estimators $\tilde{\beta}_0$ and $\tilde{\beta}_1$.)

$\hat{\beta}_0$ and $\hat{\beta}_1$ are OLS estimates of the regression:

$$Y_i = \beta_0 + \beta_1 X_i + u_i$$

$\tilde{\beta}_0$ and $\tilde{\beta}_1$ are OLS estimates of the regression:

$$\tilde{Y}_i = \beta'_0 + \beta'_1 \tilde{X}_i + \tilde{u}_i$$

Question 4

Recall that for a constant c we have $\sum_{i=1}^n cx_i = c \sum_{i=1}^n x_i$

In general, for $\tilde{X}_i = c_1 X_i$ and $\tilde{Y}_i = c_2 Y_i$, where c_1 and c_2 are constants ($c_1 \neq 0$), the following equations hold for the sample averages of $\tilde{X}_i$ and $\tilde{Y}_i, i = 1, \dots, n$ (denoted as $\bar{\tilde{X}}$ and $\bar{\tilde{Y}}$):

$$\bar{\tilde{X}} = \frac{1}{n} \sum_i^n (c_1 X_i) = c_1 \left(\frac{1}{n} \sum_i^n X_i \right) = c_1 \bar{X}$$

$$\bar{\tilde{Y}} = \frac{1}{n} \sum_i^n (c_2 Y_i) = c_2 \left(\frac{1}{n} \sum_i^n Y_i \right) = c_2 \bar{Y}$$

Question 4

Therefore, for the OLS estimator $\tilde{\beta}_1$, we have:

$$\tilde{\beta}_1 = \frac{\sum_{i=1}^n (\tilde{X}_i - \tilde{\bar{X}})(\tilde{Y}_i - \tilde{\bar{Y}})}{\sum_{i=1}^n (\tilde{X}_i - \tilde{\bar{X}})^2}$$

$$\tilde{\beta}_1 = \frac{\sum_{i=1}^n (c_1 X_i - c_1 \bar{X})(c_2 Y_i - c_2 \bar{Y})}{\sum_{i=1}^n (c_1 X_i - c_1 \bar{X})^2}$$

$$\tilde{\beta}_1 = \frac{\sum_{i=1}^n c_1 c_2 (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n c_1^2 (X_i - \bar{X})^2}$$

$$\tilde{\beta}_1 = \frac{c_1 c_2 \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{c_1^2 \sum_{i=1}^n (X_i - \bar{X})^2} = \left(\frac{c_2}{c_1} \right) \hat{\beta}_1$$

Question 4

Given $\tilde{\beta}_1 = \left(\frac{c_2}{c_1}\right) \hat{\beta}_1$, for $\tilde{\beta}_0$ we have:

$$\tilde{\beta}_0 = \tilde{\bar{Y}} - \tilde{\beta}_1 \tilde{\bar{X}} = c_2 \bar{Y} - \left(\frac{c_2}{c_1}\right) \hat{\beta}_1 \times c_1 \bar{X}$$

$$\tilde{\beta}_0 = c_2 \bar{Y} - c_2 \hat{\beta}_1 \bar{X} = c_2 (\bar{Y} - \hat{\beta}_1 \bar{X}) = c_2 \hat{\beta}_0$$

In this question, $c_1 = 10$ and $c_2 = 10$ since $\tilde{X}_i = 10 \times X_i$ and $\tilde{Y}_i = 10 \times Y_i$. Hence:

$$\tilde{\beta}_1 = \left(\frac{10}{10}\right) \hat{\beta}_1 = \hat{\beta}_1$$

$$\tilde{\beta}_0 = 10 \hat{\beta}_0$$